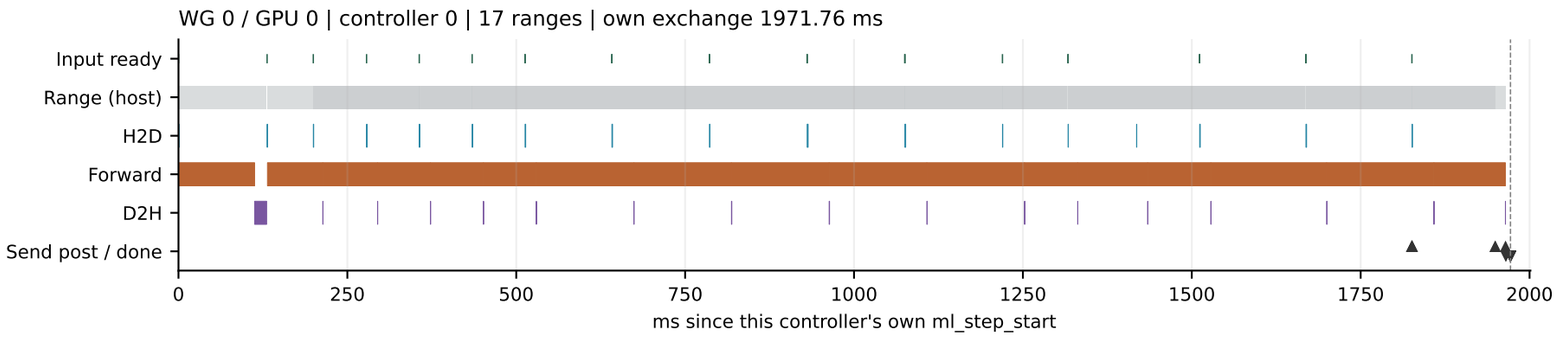
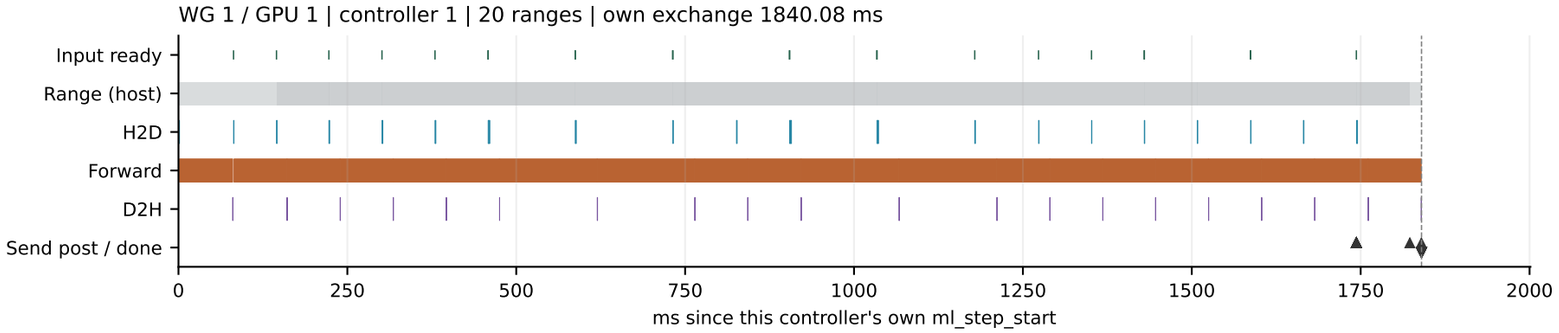
transformer | 96c4g | default | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 1719.48 ms | p2p1 2002.12 ms



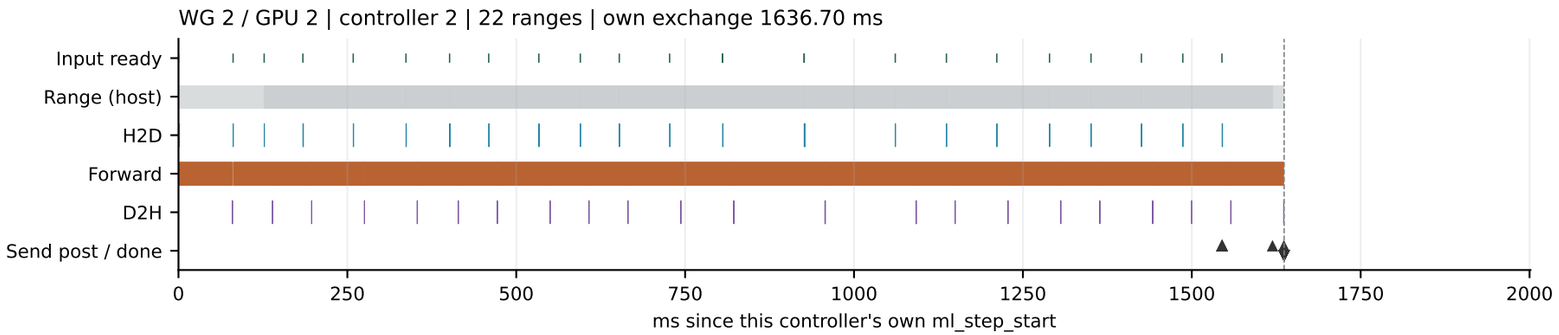
GPU work (ms)	Collective	p2p1
H2D	5.105	12.888
Forward	1615.634	1944.167
D2H	3.652	23.369
GPU SUM	1624.391	1980.424

Controllers: coll 0 / P2P 0
Coll host device elapsed: 1624.905 ms
P2P GPU envelope: 1964.315 ms
Different scopes; not an elapsed-time comparison.



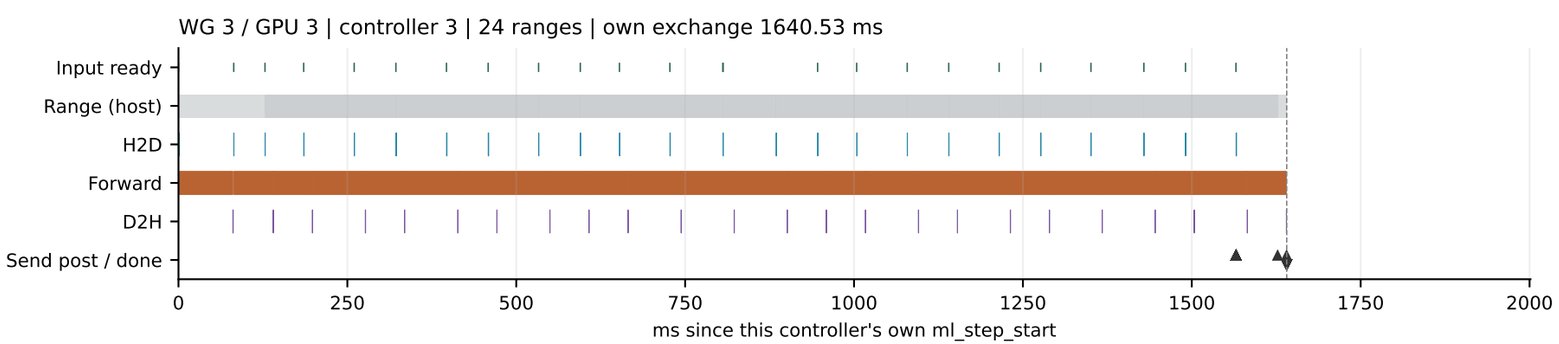
GPU work (ms)	Collective	p2p1
H2D	9.100	23.753
Forward	1615.644	1836.163
D2H	5.282	7.561
GPU SUM	1630.026	1867.477

Controllers: coll 1 / P2P 1
Coll host device elapsed: 1630.507 ms
P2P GPU envelope: 1839.685 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	4.016	11.325
Forward	1436.373	1632.974
D2H	9.097	8.419
GPU SUM	1449.485	1652.717

Controllers: coll 2 / P2P 2
Coll host device elapsed: 1450.008 ms
P2P GPU envelope: 1636.237 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	4.629	11.797
Forward	1426.385	1636.878
D2H	9.514	8.251
GPU SUM	1440.527	1656.926

Controllers: coll 3 / P2P 3
Coll host device elapsed: 1441.077 ms
P2P GPU envelope: 1640.168 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.